

Financial Sales and Loss Variance Analysis

A cross tool analytics case study using Python, SQL, R, and Power BI

Executive Summary

This analytics case study examines 700 transaction level records from the Microsoft Financial Sample dataset (September 2013 – December 2014) across five business segments, six products, and multiple countries. The analysis combines Python, SQL, R, and Power BI to investigate month-over-month (MoM) sales trends and perform root-cause analysis on profitability losses.

A critical finding emerged during initial data validation: identifying \$777,321.25 in total losses that were entirely concentrated within a single business segment and driven by specific discounting practices.

Tools and Technologies

Tool	Role in the project	Skills demonstrated
Python (pandas)	Cleaning the raw dataset and building an exploratory chart	Data validation, string parsing, handling accounting notation, date reconstruction, matplotlib visualization
SQL (SQLite)	Core aggregation, trend, and root cause analysis	GROUP BY aggregation, window functions (LAG), filtering and grouping across multiple dimensions
R	Reproducing the loss driver analysis in a second language	Base R data handling: read.csv, as.Date, subset, aggregate
Power BI	An interactive dashboard for a non-technical audience	DAX measures, KPI cards, a combo chart, and a filtered visual

Key Insights & Business Impact

- **Concentrated Loss Driver:** 100% of the 58 loss-making transactions in the dataset occurred exclusively within the Enterprise segment. All other segments (Government, Small Business, Midmarket, and Channel Partners) operated profitably.
- **Discount Policy Risk:** Losses clustered heavily in the High discount band (\$622,368.75) and Medium discount band (\$154,952.50). This pinpoints unmonitored discounting practices rather than general product or segment failure as the primary root cause.
- **Seasonality Trends:** Q4 consistently shows strong revenue surges in October (+107% MoM in 2013; +93% MoM in 2014), followed by a sharp drop in November 2014 (-56% MoM).
- **Top Geographic Market:** France generated the highest total profit (\$3,781,020.79), narrowly leading Germany and Canada.

Approach

Step 1: Data Cleaning & Validation (Python)

- **Accounting Notation Fix:** Inspected raw values and identified 58 entries formatted in accounting notation such as \$(4,533.75) representing negative values. Constructed regular expressions in Pandas to convert these into true floating-point negative numbers (-4533.75). Without this fix, \$777,321.25 in real losses would have been incorrectly summed as positive revenue.
- **Sanitization:** Used `repr()` checks to detect and strip hidden leading/trailing whitespaces across column headers and text fields. Converted currency dashes into explicit zero values.

Step 2: Relational Analytics (SQL)

- Loaded the sanitized dataset into SQLite within the notebook environment: Calculated MoM revenue, profit, and profit margin using `LAG()` window functions.
- Ran multi-dimensional `GROUP BY` queries across Segment, Country, Product, and Discount Band to pinpoint loss concentrations.

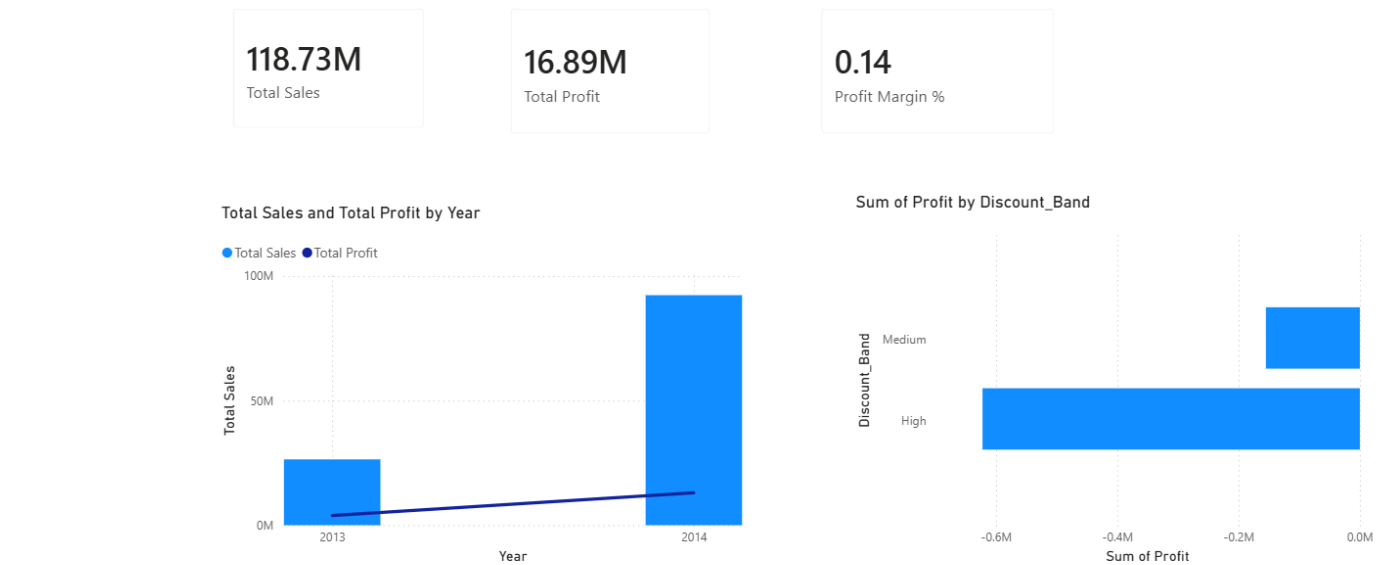
Step 3: Cross-Tool Verification (Base R)

- To verify data pipeline accuracy, the loss-driver analysis was independently replicated in Base R using read.csv(), subset(), and aggregate(). The R output matched the SQL results precisely (58 loss-making rows, \$777,321.25 total loss), confirming cross-tool data integrity.

Step 4: Visual Reporting (Power BI & Matplotlib)

- Plotted dynamic dual-axis trend lines in Matplotlib to observe seasonality despite massive scale differences between monthly sales (\$4.5M–\$12.4M) and profits (\$600K–\$2M).
- Constructed an interactive Power BI dashboard featuring DAX measures (Total Sales, Total Profit, Profit Margin %), KPI cards, a sales/profit combo chart, and a filtered visual breaking down Enterprise losses by discount band.

Financial Performance Overview



Impact and Recommendations

- **Audit Enterprise Discount Policy:** Restrict high-discount authorization rules within the Enterprise segment, as this single policy tier generates 80% of all company losses.
- **Integrate Formal Target Datasets:** Source an official budget and compare it against these results directly, rather than relying on month over month variance as a proxy.

Limitations and Assumptions

- No external budget or target data was available for this dataset, so all variance figures in this project compare each month against the previous month rather than against a formal budget.
- The dataset covers sixteen months and seven hundred rows, enough to demonstrate a full analytical workflow, but smaller than a typical production dataset.
- The Power BI dashboard summarizes the same Sales, Profit, and loss driver findings shown here, but was built directly in the application rather than from a script, so its logic is described here rather than shown as code.